CO_2 storage projects in Europe

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www.iogpeurope.org

IOGP Europe
Avenue de Tervueren 188A,
B-1150 Brussels, Belgium
T: +32 (0)2 790 7762
E: reception-europe@iogp.org

Overview of announced CO_2 storage projects in Europe

BULGARIA

1. ANRAV (IF)

CROATIA

1. Sisak Biorefinery*
2. **CO_2 EOR Project Croatia***
3. Geothermal CCS Croatia (PCI)

CZECH REPUBLIC

1. CCS Moravia

DENMARK

1. **Greensand* (FID)**
2. Bifrost* (PCI)
3. Kalundborg CCS
4. Norne (PCI)
5. Ruby
6. Greenstore*

GREECE

1. Prinos CO_2 Storage Project (PCI)

HUNGARY

1. MOL-Hungary CCS Project*
2. Danube Removals (IF)

ICELAND

1. **Orca**
2. **Silverstone** (IF)
3. Coda Terminal (IF)
4. **Mammoth**

ITALY

1. **Ravenna CCS (includes Callisto)*** (PCI)

THE NETHERLANDS

1. **Porthos* (PCI) (FID)**
2. Aramis* (PCI)
3. KOMPAS

NORWAY

1. **Sleipner***
2. **Longship (includes Northern Lights)*** (PMI)
3. **Snøhvit***
4. Smeaheia*
5. Trudvang*
6. Luna*
7. Havstjerne* (IF)
8. Poseidon (NO)*
9. Kinno*
10. Atlas*

SLOVAKIA

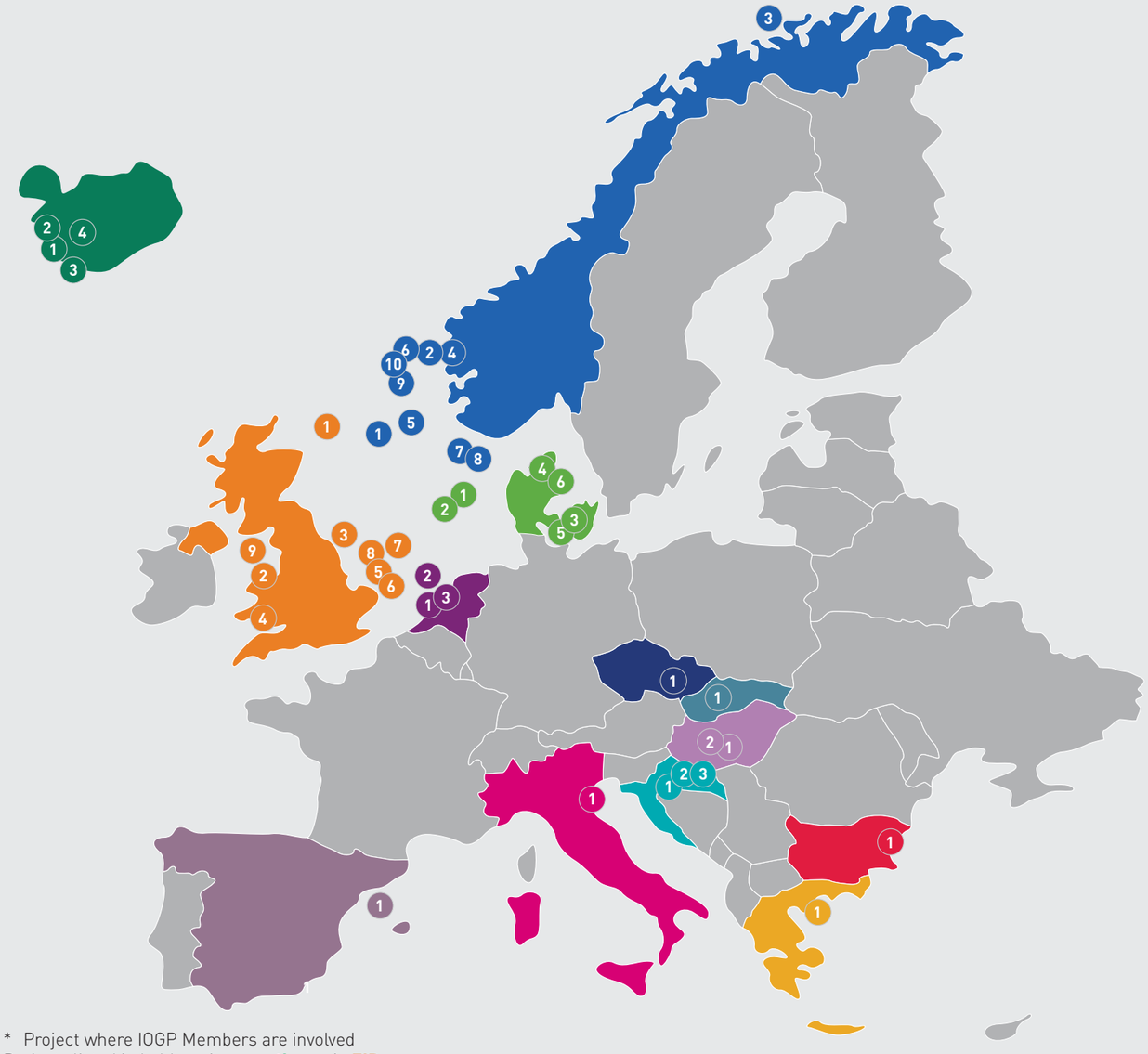
1. CarbonNet Slovakia

SPAIN

1. TarraCO2* (IF)

UK

1. Acorn*
2. **HyNet* (FID)**
3. **Net Zero Teesside* (FID)**
4. South Wales Industrial Cluster
5. Bacton Thames Net Zero*
6. Poseidon (UK)*
7. Viking CCS*
8. Orion*
9. Morecambe Net Zero Cluster



* Project where IOGP Members are involved
Projects listed in bold are in **operation** or in **FID**

(FID) – Final Investment Decision
(PCI) – Project of Common Interest
(PMI) – Project of Mutual Interest
(IF) – Project supported by the EU Innovation Fund

EU

Total projects - 20
Countries - 10
FID - 2,9 CO_2/year
Operational - 0,68 CO_2/year

Europe

Total projects - 43
Countries - 13
FID - 16,4 CO_2/year
Operational - 3,94 CO_2/year

LOCATION	NO. ON MAP	PROJECT NAME	ELEMENTS OF CCS VALUE CHAIN COVERED	TYPE OF CAPTURE PROJECT	DESCRIPTION	PARTICIPANTS	STATUS OF THE PROJECT	PLANNED START OF OPERATIONS DATE	CO_2 STORAGE INJECTION CAPACITY AT START DATE (MTPA)	CO_2 STORAGE INJECTION CAPACITY AFTER EXPANSION (MTPA)	TYPE OF CO_2 STORAGE
Bulgaria	1	ANRAV Project supported by EU Innovation Fund	Capture Transport Storage	Hard To Abate Industry (cement plant)	ANRAV aims to develop the first full CCS value chain project in Bulgaria and in Eastern Europe. The objective is to capture the CO_2 streams at the Devnya cement plant in Varna (HeidelbergCement group) and though an onshore and offshore pipeline system to store them in a depleted gas field in the Black Sea. Subject to regulatory and permitting aspects, the full-chain CCS project could be operational on 2028.	Devnya Cement (Heidelberg Cement), Petroceltic Bulgaria EOOD	Announced	2028 Expansion date 1: 2033 Expansion date 2: 2038	0.8	1st Expansion: 2 2nd Expansion: 5	Offshore - Depleted Gas Field (Galata), Black Sea
Croatia	1	Sisak Biorefinery	Capture Transport Storage	Fuel Production (Biofuels)	A pilot, full chain CCS project which is part of an advanced bioethanol production plant currently being developed at the Sisak refinery site, where advanced bioethanol from biomass will be produced. More specifically, CO_2 will be captured and transported via the existing pipeline infrastructure to be stored at the depleted oil and gas fields which are found 40 km away from the site. The project intends to capture, transport and inject 55.000 tonnes of CO_2 per year.	INA (part of MOL Group)	Announced	2027	0.055	0.3	Onshore - Depleted Oil & Gas Fields
	2	CO_2 EOR Project Croatia	Capture Transport Storage	Fuel Production (Oil & Gas)	At Gas Treatment Plant (GTP) found at Molve Municipality 640.000 m3/d of CO_2 are produced from the purification of natural gas. The CO_2 is compressed at 30 bar, dehydrated and transported via onshore pipeline 88 km long to the Fractionation Facilities of Ivanić Grad. Subsequently, the CO_2 is compressed, liquefied and transported (200 bar) for injection at the mature oil fields Ivanić and Žutica for Enhanced Oil Recovery. After years of operation, it was found out that more than 0.65 $MtCO_2$ per year is retained in the reservoirs, meaning that more CO_2 is injected than it is produced. Injection operations are deemed to continue until 2039.	MOL Group	In operation	The injection of CO_2 in Ivanić oilfield started in October 2014 and northern part of Žutica oilfields in October 2015.	0.65 *Not eligible under NZIA	no data	Onshore - Depleted Oil & Gas Fields (Ivanić & Žutica)
	3	Geothermal CCS Croatia Project of Common Interest	Capture Transport Storage	Hard to abate industry (cement plant)	The Geothermal CCS project includes the construction & repurpose of pipelines to transport CO_2 from a cement factory in Beremend (Hungary) and a cement plant in Našice (Croatia) to be stored at a saline aquifer in the north of Croatia (Bockovac site). The project includes also the construction of a geothermal plant for the production of heat and electricity where the produced CO_2 will be stored in the same saline aquifer from which the brine is extracted. The CO_2NTESSA project is linked to the Geothermal CCS project and refers to the CO_2 capture infrastructure at the cement plant in Našice (Croatia).	Croatian Hydrocarbon Agency, NEXE, INA, Plinacro, Heidelberg Materials	Announced	2029	0.63	No data	Onshore – Saline Aquifer (Bockovac site)

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Czech Republic	1	CCS Moravia	Capture Transport Storage	Hard to abate industry (cement plant)	The CCS Moravia project intends to Capture CO_2 from the Heidelberg Materials' cement plant near Brno city. The captured CO_2 will be transferred via a new pipeline to be stored in saline aquifers located in South Moravia. CO_2 storage is scheduled for 2034.The project is on a subsidy from the EU's Duty Program.	MND, Heidelberg Materials	Announced	2034	no data	no data	Onshore – Saline Aquifers, South Moravia
Denmark	1	Greensand	Capture Transport Storage	Hard to abate industry (Chemicals)	After successfully operating in its demonstration phase in March 2023, becoming the first operational full value chain CCS project in Europe, the Greensand project entered FID in 2025. Initially CO_2 was captured at the INEOS Oxide site (chemical industry) in Antwerp, Belgium, shipped cross border and stored permanently at the Danish North Sea. More specifically, 7 batches have been injected in total, equally to around 4.100 tons of CO_2. For the 1st commercial phase, 'Greensand Future', the project aspires to become operational in early 2026 with a storage capacity of 0,4 Mtpa. Finally, another expansion of the project is foreseen for up to 8 Mtpa storage capacity by 2030.	INEOS Energy, Harbour Energy, Nordsøfonden	FID	2026 Expansion date: 2030	0.4	8	Offshore - Depleted Oil & Gas Field (Nini West), Danish North Sea
	2	Bifrost Project of Common Interest	Transport Storage	Not applicable	The Bifrost project aims to transport and permantly store CO_2 in the depleted Harald gas fields by repurposing the existing offshore platform, located 250 km off the west coast of Denmark. The evaluation and appraisal study of the project (2022-2024) provides two options for the transportation of CO_2 which are both under consideration: via shipping or via a repurposed offshore pipeline. According to the work program, the 1st injection is expected by 2030 with a capacity of 3 Mtpa but expansion of the project is foreseen leading to the maximum capacity of 16 Mtpa.	TotalEnergies (45%) CarbonVault (35%) Nordsøfonden (20%)	Announced	2030 Expansion date: 2032	3	16	Offshore - Depleted Oil & Gas Field (Harald West), Danish North Sea
	3	Kalundborg CCS	Storage	Not applicable	On June 2024, the Danish authorities awarded an exploration permit for CO_2 storage to the consortium of Equinor (60%), Ørsted (20%) and Nordsøfonden (20%). The Kalundborg CCS project is located onshore, in the North West Zealand very close to the Ørsted's established shipping and storage terminal, Ørsted Kalundborg CO_2 hub. According to estimations, the project could have the potential to store up to 12 $MtCO_2$/ year when fully expanded. First injection is planned by 2030.	Equinor (60%) Ørsted (20%) Nordsøfonden (20%)	Announced	2030	no data	12	Onshore – Saline Aquifers

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Denmark	4	Norne Project of Common Interest	Transport Storage	Not applicable	Project Norne is a cross-border CO_2 transportation and storage project that aims to build dedicated onshore pipelines and a maritime network (LCO2 ship receiving facilities at Port of Aalborg and Port of Kalundborg) in order to receive CO_2 from Denmark and other countries for permanent storage into two separate onshore storage facilities (Gassum structure via Project Fyrkat and Havnsø structure via Project Trelleborg).	Fidelis New Energy, Ross Energy	Announced	2027 Expansion date: 2035	2.3	15	Onshore – Saline Aquifers (Gassum & Havnsø Structures)
	5	Ruby	Storage	Not applicable	Ruby project aims to establish an onshore CO_2 storage site on the south coast of Lolland island near Rødby town. Liquefied CO_2 will be shipped from several countries to a floating intermediate storage of CO_2, before it is injected and permanently stored onshore at Rødby Formation.	CarbonCuts (80%) Nordsøfonden (20%)	Announced	2030	1	5-10	Onshore – Saline Aquifer (Rødby Formation)
	6	Greenstore	Storage	Not applicable	On June 2024, the Danish authorities awarded an exploration permit for CO_2 storage to the consortium of Harbour Energy, INEOS Energy and the North Sea Fund. The Greenstore CCS project is located onshore, in the Gassum area (Jutland). According to estimations, the project could have the potential to store up to 5 $MtCO_2$/year when fully expanded. First injection is planned by 2029.	Harbour Energy (40%) INEOS Energy (40%) Nordsøfonden (20%)	Announced	2029	no data	5	Onshore – Saline Aquifers
Greece	1	Prinos CO_2 Storage Project Project of Common Interest	Storage	Not applicable	Prinos CO_2 Storage project will make use of the existing onshore and offshore facilities that have been producing oil & gas since 1981 from the Prinos complex. CO_2 will be received via a pipeline or via ships, stored temporarily at the onshore facilities (Sigma) and then transported via a new 20km offshore pipeline to the offshore platform (Beta) for injection and permanent storage. Phase 1 of the project will be completed on 2027 with a storage capacity of 1 Mtpa whereas Phase 2 of the project will be completed on 2029 with a storage capacity of 3 Mtpa.	Enearth (subsidiary of Energean)	Announced	2027 Expansion date: 2029	1	3	Offshore – Semi Depleted Prinos Oil Field, North Aegean Sea

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Hungary	1	MOL-Hungary CCS Project	Transport Storage	Not applicable	The project aims to implement: 1) transport of captured CO_2 to an existing MOL gas production site via a new onshore dedicated CO_2 pipeline; 2) suspension of 24 hydrocarbon production wells and the drilling of up to 8 new CO_2 injection wells, together with 8 new injector lines, associated well sites and CO_2 distribution system; 3) permanent geological storage of 0.7 MTCO2 / year.	MOL Group	Announced	No data	0.7	No expansion planned	Onshore - Depleted Oil Gas Field
	2	Danube Removals Project supported by EU Innovation Fund	Capture Transport Storage	Fuel Production (Biogas)	The Danube Removals Project intends to capture, transport and store 500.000 tons of CO_2 annually. This will happen at the lowest possible cost as the storage site is onshore (saline aquifer), the CO_2 will be transported via an onshore pipeline for a short distance of 10-15km and the CO_2 will be captured from the fermentation process (grain biorefinery) which has the world's absolute lowest costs of capture. The capex of the project is estimated at €80m and the annual opex at €14m. The CO_2 storage exploration license was granted on June 2024. Since the CO_2 will be captured from biomass-based feedstock, the project aims to generate carbon removal certificates and offset credits for the voluntary carbon market.	Danube CCS Ventures	Announced	no data	0.5	No expansion planned	Onshore – Saline Aquifer
Iceland	1	Orca	Capture Transport Storage	Direct Air Capture	Climeworks' Orca plant is the world's first and largest Direct Air Capture and mineral storage plant. Orca plant was built in 15 months, has been in operation till Sept 2021 and has the capacity of capturing and storing 4.000 tons of CO_2 per year. More specifically, the CO_2 which is captured from the atmospheric air is mixed with water and injected into nearby basaltic rock-formations to be permanently turned into stone.	Climeworks, Carbfix	In operation	In operation	0.004	no data	Onshore - Basalt rock formations
	2	Silverstone Project supported by EU Innovation Fund	Capture Transport Storage	Power Production (Geothermal Energy)	Located at the Hellisheiði geothermal power plant in Iceland, the Silverstone project applies Carbfix technology to capture and permanently store CO_2 by mineralising it in basaltic rock formations. With a capture capacity of 34,000 tonnes of CO_2 per year, Silverstone builds on earlier pilot facilities (12,000 tCO_2/year) and will remove a total of around 150,000 tonnes of CO_2 during its operational lifetime. The project, developed by Carbfix and ON Power and co-funded by the EU Innovation Fund, is expected to deliver about 10% of Iceland's planned emission reductions under its 2030 Climate Plan.	Carbfix, ON Power	In operation	In operation	0.012	0.034	Onshore - Basalt rock formations

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	3	Coda Terminal Project supported by EU Innovation Fund	Transport Storage	Not applicable	Coda Terminal is Iceland's first large-scale, cross-border CO_2 transport and mineral storage hub at Straumsvík. It is now slated to enter operation in 2029, with the initial phase ramping up to approximately 0.7 $MtCO_2$ per year before ultimately achieving a full capacity of 3 $MtCO_2$ annually by 2032. Captured CO_2 from Northern European industrial sources will be shipped to Iceland, unloaded into onshore tanks, then dissolved in water and injected via a network of wells into the basaltic bedrock, where it mineralises into stable carbonate rock within two years using Carbfix technology. Over its first decade of operation, the terminal is expected to avoid approximately 21 $MtCO_2$ equivalent emissions, which equates to over half of Iceland's annual emissions and around 2.5% of the EU-wide emissions reductions needed by 2030.	Carbfix, Dan-Unity CO_2 A/S	Announced	2029 Expansion date: 2032	0.7	3	Onshore - Basalt rock formations
	4	Mammoth	Capture Transport Storage	Direct Air Capture	Mammoth is a Direct Air Capture project which entered in operation in May 2024. The project has a CO_2 capture capacity of 36.000 tons per year. The CO_2 captured is mixed with water to be injected underground for mineral storage in basaltic rocks.	Climeworks, Carbfix	In operation	2024	0.036	no data	Onshore - Basalt rock formations
Italy	1	Ravenna CCS (includes Callisto) Project of Common Interest	Capture Transport Storage	Cluster (multiple industrial sources)	The project involves the establishment of a CCUS hub in Ravenna with the perspective to capture CO_2 from multiple industries in Italy and abroad and store it in the depleted gas fields of the Adriatic Sea. Phase 1 of the project was launched in September 2024. Since then, CO_2 emissions (25,000 t per year) have been captured at Eni's natural gas treatment plant in Casalborsetti, transported via repurposed gas pipelines and stored offshore at the depleted gas field Porto Corsini Mare Ovest. Phase 2 of the project is now scheduled for 2030 with a capacity of 4 Mtpa. The project also retains significant expansion potential, with storage capacity expected to increase progressively to 12 Mtpa by 2034 and to 16 Mtpa by 2038.	Eni CCS Holdings, Snam	In operation	In operation Expansion date: 2030	0.025 *Eligible under NZIA	4	Offshore – Depleted Gas field (Porto Corsini Mare Ovest), Adriatic Sea (off the coast of Ravenna)

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The Netherlands	1	Porthos Project of Common Interest	Transport Storage	Not applicable	Porthos stands for Port of Rotterdam CO_2 Transport Hub and Offshore Storage. The project intends to provide transport and storage infrastructure to energy intensive industries in the Port of Rotterdam and possibly at a later stage to industries in the Antwerp and North Rhine Westphalia areas. The project will link the CO_2 Capture facilities and the existing OCAP pipeline with a new onshore pipeline which will drive the aggregated CO_2 in a CO^2 hub in the Port of Rotterdam and subsequently via an offshore pipeline in a depleted gas field 20 km off the coast for permanent storage. FID was taken in Oct 2023.	EBN, Gasunie, Port of Rotterdam Authority	FID	2027	2.5	no data	Offshore - Depleted Gas Fields (P18-2, P18-4 and P18-6), Dutch North Sea
	2	Aramis Project of Common Interest	Transport Storage	Not applicable	Aramis project aspires to enable multiple energy intensive industries to store their captured CO_2 emissions under the Dutch North Sea. For this purpose, a CO_2 collection hub and an offshore pipeline with a 22 Mt/year transport capacity will be built to connect the port of Rotterdam with the depleted gas fields located 200 km off the coast. Industries will be able to connect with the CO_2 collection hub either by onshore pipelines either by ships. The FEED phase was finalized on Nov 2024.	TotalEnergies, Shell, EBN, Gasunie	Announced	2030 Expansion date: 2035	5	22	Offshore - Depleted Gas Field (L7) and potential Aquifers, Dutch North Sea
	3	KOMPAS	Storage	Not applicable	Kompas is a storage project developed by ONE-Dyas and EBN to permanently store captured CO_2 emissions from hard-to-abate industries in the depleted Q16-FA gas field in the Dutch sector of the North Sea, approximately 10 km north of the Maasvlakte. Captured CO_2 will be transported by ship to an offshore unloading station and injected through the Q16-B platform into the depleted reservoir beneath the seabed. The project is expected to provide a total storage capacity of around 15 Mt of CO_2.	ONE-Dyas & EBN	Announced	2030	1	no data	Depleted Q16-FA gas field
Norway	1	Sleipner	Capture Transport Storage	Upsteam Oil & Gas (Gas Processing)	Sleipner CCS project has been operational since 1996 and it is known as the world's first offshore CCS implementation. The produced natural gas at Sleipner area contains around 9% of CO_2 and therefore the production is driven to a processing and CO_2 removal platform (Sleipner T) where 2.800 tonnes of CO_2 are separated and injected daily in the Utsira saline formation, 800m below the seabed. By the end of 2020 more than 19 Mt of CO_2 have been injected and permanently stored. The project is the result of a tax on CO_2 emissions which was introduced in Norway in 1991.	Equinor, ExxonMobil, Total	In operation	In operation since 1996	1	no data	Offshore - Saline Aquifer (Utsira Formation), Norwegian North Sea

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Norway	2	Longship (includes Northern Lights 1) Project of Mutual Interest	Capture Transport Storage	Cluster (multiple industrial sources)	Longship CCS project includes the capture of 0.4 Mtpa of CO_2 from a cement plant in Brevik and 0.4 Mtpa of CO_2 from a waste incineration plant in Oslo along with the liquefaction and the temporary storage of the captured CO_2 at the premises of the industries. The Northern Lights project is part of the Longship CCS project and involves the shipping of the liquefied CO_2 and its permanent storage at the Aurora saline aquifer below the Norwegian North Sea. Phase 1 of Northern Lights project has been operational since Sept 2024, being able to transport & store 1,5 Mtpa of CO_2. Northern Lights has the vision to expand and receive additional volumes of CO_2 from several industries in Europe.	Equinor (33,3%), Shell (33,3%), Total Energies (33,3%)	In operation	In operation (since 2024)	1.5	Expansion through Northern Lights 2 project	Offshore - Saline Aquifer (Aurora), Norwegian North Sea CO_2 License: EL 001
		Northern Lights 2	Transport Storage	Not applicable	The Northern Lights project is part of the Longship CCS project and involves only the transport and storage part. Northern Lights 1 has been in operation since Sept 2024 with a transport and storage capacity of 1.5 Mtpa. Northern Lights 2 intends to have a transport and storage capacity of 3.5 Mtpa by 2028, dependent on market demand.	Equinor (33,3%), Shell (33,3%), Total Energies (33,3%)	FID	2028	3.5	no data	Offshore - Saline Aquifer (Aurora), Norwegian North Sea CO_2 License: EL 001
	3	Snøhvit	Capture Transport Storage	Fuel Production (Oil & Gas)	Snøhvit is an offshore gas field in Northern Norway, that supplies gas to the Melkøya LNG production facility. Since the produced natural gas contains 5-6% CO_2, it is separated and the CO_2 is piped back to the Snøhvit field. There the CO_2 is injected for permanent storage into the Tubaen saline formation at 2600m depth, beneath the gas production layers. The CCS project is the result of a tax on CO_2 emissions which was introduced in Norway in 1991.	Equinor	In operation	In operation since 2008	0.7	no data	Offshore - Saline Aquifer (Tubaen), Barents Sea
	4	Smeaheia	Transport Storage	Not applicable	The Smeaheia CCS project aims to transport and store 5 Mtpa of CO_2 by the early 2030s. Two transport concepts are being considered to connect North West Europe with the storage site: A pipeline solution (CO_2 Highway Europe) and a shipping solution coupled with an onshore receiving terminal on the west coast of Norway (Smeaheia Norwegian Hub). The project could expand and store up to 20 Mtpa by 2035.	Equinor	Announced	Early 2030s Expansion date: 2035	5	20	Offshore - Saline Aquifer (Sognefjord), Norwegian North Sea CO_2 License: EXL 002
	5	Trudvang	Transport Storage	Not applicable	The Trudvang project aims to receive liquifed CO_2 via shipping to an onshore terminal in the south-west of Norway and transport it via a purpose-built pipeline to the Norwegian North Sea (about 200 kilometres from the coast) for injection and permanent storage. Technical feasibility of the project has been completed on Aug 2024.	Storegga (30%), Var Energi (40%), INPEX (30%)	Announced	2029	9	no data	Offshore - Saline Aquifer (Utsira), Norwegian North Sea CO_2 License: EXL 007

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Norway	6	Luna	Storage	Not applicable	Luna CCS project is located in the Norwegian North Sea, 120 km offshore Bergen. The project aims to store up to 5 $MtCO_2$/year by 2030. Technical feasibility of the project has been completed on Aug 2024.	Habour Energy (60%), TotalEnergies (40%)	Announced	2030	5	no data	Offshore - Saline Aquifer (Statfjord), Norwegian North Sea CO_2 License: EXL 004
	7	Havstjerne Project supported by EU Innovation Fund	Storage	Not applicable	Havstjerne is an offshore open access CO_2 storage project that aims to permanently store liquefied CO_2 that will be shipped from several CO_2 hubs across Europe. For this purpose, the STARFISH project (Sequestration Technology And Reservoir: Floating Injection and Storage in Havstjerne) has been awarded a grant from the EU's Innovation Fund. The STARFISH project will feature a specially designed injection unit capable of directly receiving large volumes of liquefied CO_2 from purpose-built LCO2 carriers. Technical feasibility of the project has been completed on Aug 2024.	Habour Energy (60%) Stella Maris CCS (40%)	Announced	2030 Expansion date: 2033	3	7	Offshore - Saline Aquifer (Sandnes), Norwegian North Sea CO_2 License: EXL 006
	8	Poseidon (NO)	Storage	Not applicable	Poseidon CCS license is found in the Norwegian North Sea, about 100 km off the Norwegian coast. In Mar 2023, Aker BP and OMV were awarded the Poseidon license and in Oct 2023 the acquisition of the 3D seismic data has been completed. According to the work program, FID is expected before May 2025. The Poseidon CCS project could potentially provide a storage injection capacity of more than 5 Mtpa of CO_2 that will be captured mainly from industrial emitters in North-West Europe.	Aker BP (50%), OMV (50%)	Announced	2030	5 Mtpa	No data	Offshore - Norwegian North Sea CO_2 License: EXL 005
	9	Kinno	Storage	Not applicable	On June 2024, the Norwegian Ministry of Energy awarded an exploration permit for CO_2 storage to Equinor. The Kinno CCS project is located in the Norwegian North Sea, 130 km west of the coast from Haugesund, it is estimated that has the potential to store up to 5 $MtCO_2$/year. According to the work program of the exploration permit, Equinor should make a FID in 4 years (Jun 2028). The Kinno project will be one of the storage sites to be connected with the planed 1000km long pipeline CO_2 Highway Europe in order to receive CO_2 from European industrial emitters.	Equinor	Announced	no data	no data	5	Offshore – Norwegian North Sea CO_2 License: EXL 010

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Norway	10	Atlas	Storage	Not applicable	On June 2024, the Norwegian Ministry of Energy awarded an exploration permit for CO_2 storage to the consortium of Aker BP, PGNiG Upstream, Norway AS. The Atlas CCS project is located in the Norwegian North Sea, 130 km west of the coast from Haugesund, very close to the Yggdrasil field. According to the work program of the exploration permit, the consortium should make a FID in 4 years (Jun 2028).	Aker BP (80%) Orlen Upstream Norway (20%)	Announced	2030	6	no data	Offshore – Norwegian North Sea CO_2 License: EXL 011
Slovakia	1	CarbonNet Slovakia	Transport Storage	Not applicable	Engas aims to convert an Underground Gas Storage site operating the last 25 years at the Golianovo field located 70 km North East of Bratislava into the 1st onshore CO_2 Storage site in Slovakia. Over time the UGS site lost its strategic attractiveness and now the existing infrastructure will be repurposed for CO_2 storage. Engas plans the 1st injection by 2027 with a capacity of 350.000 t/year.	Engas	Announced	2027 Expansion date: 2033	0.3	0.5	Onshore – Depleted Gas Field (Golianovo field)
Spain	1	TarraCO2 Project supported by EU Innovation Fund	Storage	Hard to abate (petrochemical; cement; other)	The TarraCO_2 project, led by Repsol, is Spain's first commercial-scale carbon capture and storage (CCS) initiative. It is designed to capture CO_2 emissions from industrial facilities - primarily in the chemical and cement sectors of the Tarragona region - and transport them via pipeline for storage located 45 km offshore in the Mediterranean Sea. The project aims to store up to 2 million tonnes of CO_2 per year, with a total capacity of 54 million tonnes, contributing to a significant reduction in regional industrial emissions. It is intended to serve as a foundation for a future CO_2 storage hub, potentially expanding to include other emitters and carbon removal technologies. The initiative has received €205 million in EU Innovation Fund support and is planned to be operational by 2030.	Repsol	Announced	2030	2 Mtpa	N/A	Offshore
UK	1	Acorn	Capture Transport Storage	Fuel Production (Hydrogen)	Acorn is a joint venture of companies which is considered to be the backbone of the Scottish Cluster. Acorn is working on three linked projects: Acorn Carbon Capture project that aims at the construction of a carbon capture plant for the decarbonisation of two St. Fergus Gas Terminals. Acorn Hydrogen project that aims at the construction of a Blue Hydrogen facility adjacent to the two St. Fergus Gas Terminals. Acorn Transport & Storage project that aims at the transportation of the captured CO_2 for permanent storage under the North Sea using repurposed pipelines and ships.	Storegga, Shell, Harbour Energy and NSMP	Announced	2032	5	N/A	Offshore - Depleated Gas Field (Goldeneye Field), British North Sea

LOCATION	NO. ON MAP	PROJECT NAME	ELEMENTS OF CCS VALUE CHAIN COVERED	TYPE OF CAPTURE PROJECT	DESCRIPTION	PARTICIPANTS	STATUS OF THE PROJECT	PLANNED START OF OPERATIONS DATE	CO_2 STORAGE INJECTION CAPACITY AT START DATE (MTPA)	CO_2 STORAGE INJECTION CAPACITY AFTER EXPANSION (MTPA)	TYPE OF CO_2 STORAGE
UK	2	HyNet North West	Capture Transport Storage	Cluster (multiple industrial sources)	HyNet North West project aspires to decarbonise the North West Cluster. The project aims to develop the infrastructure for the production, distribution and storage of low carbon hydrogen along with the infrastructure for the capture, transport and permanent storage of the CO_2 emissions coming from the industry. The captured CO_2 will be stored at the depleted Hamilton gas field in Liverpool Bay. Eni is the developer and operator of the Liverpool Bay CO_2 Transport and Storage network as part of the HyNet Cluster. CO_2 injection readiness: 2028.	Progressive Energy, Cadent, CF Fertilisers, Eni UK, Essar, Hanson, INOVYN (part of the INEOS Group) and the University of Chester.	FID	2028 Expansion date: 2030	4.5	10	Offshore - Depleted Gas Field (Hamilton), Liverpool Bay
	3	Net Zero Teeside	Capture Transport Storage	Cluster (multiple industrial sources)	Net Zero Teesside (NZT) along with the Zero Carbon Humber (ZCH) and the Northern Endurance Partnership (NEP) constitute the East Coast cluster which is the most carbon intensive industrial cluster in UK. The following three Net Zero Teesside projects: Net Zero Teesside Power, H2Teesside and Teesside Hydrogen CO_2 Capture will provide 4 Mtpa of CO_2 by 2028. Northern Endurance Partnership (NEP) which BP leads as operator, will provide the infrastructure needed to transport (via an offshore pipeline 145 km) and permanently store the captured CO_2 at the Endurance aquifer (North Sea).	BP, Eni, Equinor, Shell, Total, Teeside Valley Authority, Suez, BOC, sembcorp,CF, Lotte Chemical, Nepic, UK BEIS	FID	2028	4	no data	Offshore - Saline Aquifer (Endurance), British North Sea
	4	South Wales Industrial Cluster	Capture Transport Storage	Cluster (multiple industrial sources)	Industries in the South Wales region produce 16Mtpa of CO_2. The South Wales Industrial Cluster which was formed in 2019, consists of major industries along the M4 corridor. Currently, two projects Cluster Plan and Deployment Project aim to shape the strategy for the decarbonization of the South Wales Industrial Cluster. The potential for offshore storage around Wales is low but new offshore sites could be investigated. Alternatively, captured CO_2 will need to be transported via shipping to existing storage sites outside Wales.	RWE, Progressive Energy, Cadent, Valero, Tata Steel, BOC, South Hook, Wales & West Utilities, Western Power Distribution, Rice, Flexis	Announced	no data	no data	no data	Offshore - Depleted Oil & Gas Fields & Saline Aquifers, Celtic Sea and Bristol Channel
	5	Bacton Thames Net Zero	Transport Storage	Not applicable	The Bacton Thames Net Zero initiative aims at the convention of the Hewett depleted gas field located 20 km offshore from Bacton (Southern North Sea), to a CO_2 Storage. It is estimated that the Hewett field could have a total CO_2 storage capacity of 330Mt. On Sep 2023, Eni has been awarded the CO_2 Appraisal and Storage Licence from NSTA. The appraisal term will last 5 years and FID is expected around 2027. First injection readiness is expected before 2030. The CO_2 storage injection capacity will be around 6 Mtpa by 2030 but expansion is foreseen for 10 Mtpa on a later stage (2035).	ENI	Announced	2030 Expansion date: 2035	6	10	Offshore - Depleted Gas Field (Hewett), British North Sea CO_2 License: CS008

LOCATION	NO. ON MAP	PROJECT NAME	ELEMENTS OF CCS VALUE CHAIN COVERED	TYPE OF CAPTURE PROJECT	DESCRIPTION	PARTICIPANTS	STATUS OF THE PROJECT	PLANNED START OF OPERATIONS DATE	CO_2 STORAGE INJECTION CAPACITY AT START DATE (MTPA)	CO_2 STORAGE INJECTION CAPACITY AFTER EXPANSION (MTPA)	TYPE OF CO_2 STORAGE
UK	6	Poseidon (UK)	Transport Storage	Not applicable	The Poseidon project involves the transportation of CO_2 (received from several industrial sources) from the Bacton terminal to the depleted Leman gas field via an offshore pipeline for permanent storage. The project will start operating in 2029 with 1.5 Mtpa injection capacity whereas it is foreseen to increase the injection capacity at 10 Mtpa by 2034. On Feb 2025, UK's first-ever CO_2 injection test took place. FID target is for 2028.	Perenco, Carbon Catalyst, Harbour Energy	Announced	2029 Expansion date: 2034	1.5	10	Offshore - Depleted Gas Field (Leman gas field) & Saline Aquifers, British North Sea CO_2 License: CS009
	7	Viking CCS	Capture Transport Storage	Cluster (multiple industrial sources)	The Viking CCS Cluster involves: 1) CO_2 capture from several industries in the Humber, Lincolnshire and Nottinghamshire regions, 2) CO_2 transportation from the Theddlethorpe Gas Terminal via an existing offshore pipeline, 3) CO_2 storage at the depleted Viking gas fields (North Sea). The Cluster is targeting to store 10 Mtpa by 2030, rising to 15 Mtpa by 2035.	Harbour Energy, bp, Phillips 66 Limited, VPI, Associated British Ports (ABP), West Burton Energy, RWE	Announced	2030 Expansion date: 2035	4	10	Offshore - Depleted Gas Field (Viking field), British North Sea
	8	Orion	Transport Storage	Not applicable	The Orion project involves the transportation of CO_2 (received from several industrial sources) from the Dimlington terminal to the depleted Amethyst gas field and to the West Sole field via offshore pipelines for permanent storage. The project will start operating in 2031 with 1 Mtpa injection capacity whereas it is foreseen to increase the injection capacity at 6 Mtpa at latter stages (2035). FID is expected in 2029.	Perenco, Carbon Catalyst	Announced	2031 Expansion date: 2035	1	6	Offshore - Depleted Gas Field (Amethyst field and West Sole field), British North Sea CO_2 Licenses: CS017 & CS018
	9	Morecambe Net Zero Cluster	Transport Storage	Not applicable	On May 2023, the North Sea Transition Authority awarded an exploration permit for CO_2 storage to Spirit Energy related to the depleted North & South Morecambe gas fields, in operation since 1974 and located in the East Irish Sea off northwest England. The project aims to accept CO_2 at first by a repurposed pipeline and on a later basis by ship and rail. FID is expected by 2028 whereas first injection by 2030.	Spirit Energy	Announced	2030	no data	no data	Offshore – Depleted Gas Fields (North & South Morecambe gas fields)

Carbon Capture, and Storage

CCS is a set of technologies that enable the Capture, Transport and Storage of CO_2.

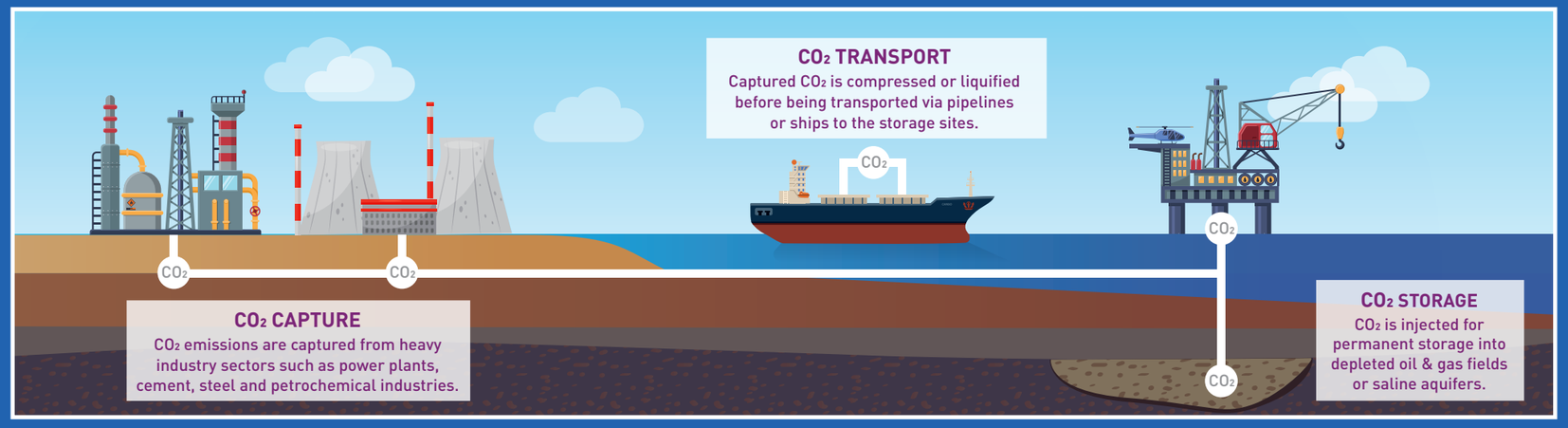
CCS is a proven and safe technology. CO_2 has been captured, transported and stored in Europe successfully since 1996 (Sleipner project, Norway).

It is a key technology for Europe to meet climate neutrality.

More CCS resources at iogpeurope.org

How it works

The 3 segments of the CCS value chain



CCS be deployed at scale, often repurposing existing infrastructures

Where can CCS make a difference?

Decarbonisation of hard-to-abate industries

In the EU, steel, cement, chemical and refining sectors emit 37% of total CO_2 industrial emissions. CCS is one of the only technological options to enable emission reductions in hard-to-abate industries.

Energy transition

CCS can be applied to gas-fired power plants which provide flexibility to an electricity grid with a higher share of intermittent renewables.

Low carbon hydrogen production

Hydrogen production based on natural gas decarbonized with CCS is the most cost-effective. It can supply industrial sectors and decarbonize sectors which cannot be electrified such as aviation and maritime transport.

Negative emissions

Large scale negative emissions can be achieved when BioEnergy production is combined with CCS (BECCS) or when Direct Air Capture is combined with CCS (DACCS).

International Association of Oil & Gas Producers

IOGP Europe
Avenue de Tervueren 188A, B-1150 Brussels, Belgium
T: +32 2 882 16 53
E. reception-europe@iogp.org

IOGP Headquarters
3 Moorgate Place, London, EC2R 6EA, United Kingdom
T: +44 20 4570 6879
E: reception@iogp.org

www.iogpeurope.org
www.iogp.org